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How do small-holder farmers understand insurance, and how much do they want it? Evidence from Africa

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ABSTRACT

Innovative micro-insurance schemes can be one tool to help African small-holder farmers cope with climate variability and change. A critical challenge for implementing such risk transfer programs, however, is helping participants understand how insurance operates, and there is evidence that farmers with a poor understanding of insurance are less likely to use it. One of the proposed tools to help farmers understand insurance is a simulation game, through which farmers can gain first-hand experience with a functioning insurance market. This paper reports on the results of experiments in Ethiopia and Malawi, investigating farmers' understanding and the effectiveness of a role-playing game at improving that understanding. Our results suggest a generally poor understanding of basic insurance concepts, and are consistent with past results in suggesting that better understanding correlates with greater willingness to purchase it. Our results also suggest that role-playing games may be an important tool for improving understanding, but that they do not necessarily out-perform more conventional training practices.

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1. Introduction

One of the major challenges facing the majority of Africans is coping with interannual rainfall variability. In sub-Saharan Africa, 200 million people go hungry in years of poor rainfall (IRI, 2005). The agricultural sector accounts for 70% of total employment, roughly 40% of exports, and 35% of Gross Domestic Product. Most of this agriculture exists without access to irrigation; over 95% of the crop-land in Africa is devoted to rain-fed agriculture, and over 90% of the population depends on rain-fed agriculture for their basic food requirements (FAO, 2000). The inability to manage risks associated with climate variability in Africa constitutes an impediment to human development, and to progress towards the Millennium Development Goals (MDGs). Given projections that climate variability – in terms of both extreme events and seasonal anomalies – is likely to increase, coping with climate variability is also a critical concern for climate change adaptation (Boko et al., 2007; Washington et al., 2006).

In a landmark study, Scoones et al. (1996) documented how dryland subsistence farmers in Zimbabwe used a diverse set of strategies to cope with a complex set of risks. As the researchers documented, these made use of both traditional knowledge and practice and modern farming methods, and also revealed the inter-dependencies between these farmers and their relatives who had migrated to cities, who often provided remittances. Since then, there has been a continued effort both to document farmers' coping strategies, and to test ways of augmenting their capacity (Thomas et al., 2007; Ziervogel and Calder, 2003). A large number of researchers have examined the extent to which seasonal climate forecasts can be used by farmers to prepare for both good and bad years (IRICP, 2000; Klopper, 1999; Klopper et al., 2006; Orlove and Tosteson, 1999; Patt et al., 2007; Ziervogel, 2004). While there have been several documented cases of disappointment (Broad and Agrawala, 2000; Tarhule and Lamb, 2003), more recent results from West Africa (Diarra and Kangah, 2007; Roncoli et al., 2009) and southern Africa (Patt et al., 2005) have demonstrated the benefits that these can bring through actual field trials. Other research has examined how responses are mediated by farmers' perceptions of climate variability and risk (Orlove and Kabugo, 2005), or farmers' asset base and their capacity to buffer risk (Osbahr et al., 2008). Likewise in West Africa, for example, Roncoli et al. (2001) showed how both formal and informal risk spreading measures were used to manage variability.

One of the common themes across this research, however, has been that many coping strategies are challenged in successive years of particularly poor rainfall (White et al., 2005), when entire communities are hit by the a common event that makes it impossible to provide relief through traditional safety nets, and often challenges international relief efforts (Linnerooth-Bayer

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et al., 2005). For this reason, there has been growing interest within the development community to explore the possibility of agricultural insurance tailored to the needs of African smallholder farmers (Osgood and Warren, 2007). At a most basic level, such insurance can draw on outside capital reserves to provide relief (Hochrainer et al., 2009; Meze-Hausken et al., 2009). Such insurance can also, however, provide benefits that allow farmers to improve their average harvests, such as by guaranteeing the repayment of loans, which in turn would allow farmers to plant higher yielding crop varieties (Suarez et al., 2007).

In many other cases dealing with ways that African smallholder farmers can cope with climate variability, an important barrier to the uptake of new methods or information is the lack of understanding and trust (Cash et al., 2006). This difficulty has been likewise demonstrated in the case of index insurance, where several studies have shown that a lack of understanding and trust has played a role in reducing the effectiveness of insurance programs (Churchill, 2006; Cohen and Sebstad, 2006; Dercon et al., 2008; Leftley and Mapfumo, 2006).

Researchers have, with generally good results, examined a variety of communication and decision-support systems to overcome these barriers, including the promotion of stakeholder networks (Ziervogel and Downing, 2004), discussion groups (Phillips and Orlove, 2004), expert workshops (Patt and Gwata, 2002; Patt et al., 2005; Roncoli et al., 2009), and role-playing games (Marx et al., 2007). It is clear that many of these methods are successful or not depending on a large number of contextual factors, such as the institutional relationship between farmers and the organizations involved (Cash et al., 2003; Guston, 2001), or farmers' perceptions of how risk levels are changing (Patt and Schröter, 2008).

In this paper we report on an experiment conducted with farmers in Ethiopia and Malawi, the purpose of which was to test whether games could be an effective way to communicate the concepts of index insurance, making them useful as a decision-support tool. The results we report address three related questions. First, to what extent do farmers understand innovative insurance tools as a means to reduce the risk associated with climate variability? Second, does a communication strategy built around the use of simulation games improve understanding, more than the more conventional approach of simply explaining how insurance works? Third, to what extent does farmers' desire to purchase insurance correlate with their level of understanding?

2. Background

In most developed countries, insurance plays an important role in protecting farmers from risk, and in so doing allowing them to make decisions that bring higher average returns. The basic concept of insurance is to spread the risk of loss over many years, and across a wide population base. In every year, participants pay a premium that they can afford, and then in years of suffering a loss, they collect a larger payout, which protects them from insolvency or livelihood insecurity. In an actuarially fair insurance contract, the premium is exactly equal to the average, or expected, payout, while in a heavily subsidized insurance contract the premium may be less than the average payout. In most insurance contracts, by contrast, the premium is more than the expected payout, and it is in this way that insurance companies cover their administrative expenses, pay for the cost of holding capital reserves, and earn a profit for their shareholders. Farmers are nevertheless willing to purchase such insurance for two reasons. First, they are typically risk averse, which means that the consequences of relatively infrequent large losses – which the insurance contract reduces – are much more serious to them than the effects of having to pay the relatively small premium each year. Second, insurance may often be a precondition for borrowing money, in order to protect the lender from the consequences of the farmers' bad luck. Many farmers borrow money each year to pay for their farming inputs, such as hybrid seeds and fertilizer, and then pay off the loan with proceeds from the harvest.

The same rationale for purchasing insurance applies in developing countries as well (Freeman and Kunreuther, 2002). However, in developing countries the total value insured is often quite small, while the administrative expenses for the insurance company are often at least as large as in developed countries, since it is time consuming and expensive to visit remote farmers' fields in order to verify actual losses. As a result insurance premiums, relative to the value insured, need to be so high that they may be unaffordable to most farmers. One way of substantially reducing administrative costs is through the use of index insurance. Payouts from index insurance contracts are not tied to actual verified losses, but rather to the occurrence of an event that causes loss, as measured by a predefined index, such as rainfall measured at a nearby weather station during the germination period. Farmers receive payouts if and only if the index falls below a certain value, with this value being chosen to match, as well as possible, the likelihood that farmers' crops will have been damaged. While it can be made far more affordable than conventional insurance, index insurance has the drawback of not covering the risk completely. There is some chance, known as basis risk, that a given farmer will suffer a poor harvest even though the value of the index is high, and as a result not be compensated for the loss.

The presence of basis risk makes even more complicated a topic that is challenging to begin with, especially for smallholder farmers with little education and even less background in insurance. Johnson et al. (1993) showed that many people poorly estimate the probability of loss, confounding their ability to evaluate whether a given insurance contract is fair for them or not. Of particular importance for index insurance is the observation of probability weighting, by which people consistently overestimate the likelihood of extremely rare events (Camerer, 2001; Kahneman and Tversky, 1979; Tversky and Kahneman, 1992). This could in turn cause people to underestimate the difference between their ordinary risk of loss and their level of basis risk, and hence view index insurance as providing them with little benefit. There is substantial evidence that people learn to make these kind of difficult decisions better over time, as a consequence of evolving heuristics (Gigerenzer and Selten, 2001; Nisbett et al., 1983; Payne et al., 1993; Simon, 1956), and that an understanding of probabilities and their importance to a particular decision, gained through received information rather than through personal experience, is no substitute (Weber, 2006; Weber et al., 2000).

A study by Giné and Yang (2009) provides an example of these difficulties. The researchers reported on the result of a randomized field experiment conducted in Malawi that concerned the decision to purchase insurance. The insurance contract there was tied to a loan to be used to purchase high-yielding farming seeds. One group of farmers was given the opportunity to take a loan without insurance, while a second group of farmers was given the opportunity to take a loan that was bundled to an index insurance contract, which would repay the value of the loan should rainfall prove poor. Interestingly, a lower proportion of farmers (20% compared to 33%) took the loan when it was bundled with insurance, suggesting that something about the insurance was unattractive. However, take-up of the insured loan was significantly higher among better-educated farmers, while there was no difference observed in take-up of the uninsured loan according to education. This then suggests that the unattractive feature of the insurance may have been tied to whether farmers fully understood what insurance was.

In the absence of direct personal experience with a particular decision-making problem, such as whether to purchase index insurance, it may be possible to provide people with the opportunity for experiential learning through the use of simulation games, capturing the essential feature of the problem (Krolikowska et al., 2007; Martin et al., 2007). For example, to assist farmers make decisions for managing a complex watershed in Senegal, Barreteau et al. (2001) relied on a role-playing game that mimicked the essential features of the multi-agent system. In Zimbabwe, Suarez and Patt (2004) used a game that involved betting on the outcome of a football game in order to teach the same concepts of conditional probability that would be necessary to understand the sources of uncertainty in a seasonal climate forecast. Games can also be an effective tool to investigate how people are likely to behave in a particular decision-making context, and whether particular changes to the decision-making problem affect that behavior (Kagel et al., 1995). In another example involving Zimbabwe, Patt et al. (2006) employed a probabilistic decision-making game to examine the factors that enhanced or detracted from farmers' trust in expert advice.

A number of researchers have started to use insurance-related games both as an educational and as a research tool (Patt et al., 2009). Carter et al. (2008) report on games that they developed and administered among cotton farmers in Peru and herders in Kenya. In both games, they attempted to tailor the games as closely as possible to actual market conditions. Participants began the game with a certain level of productive assets, and then could decide whether to take a loan or to purchase insurance. The games simulated both the performance of the index and the presence of basis risk through the use of two colored chips. A single chip representing the rainfall index was drawn for all the participants for each of the simulated years, while each participant individually drew a second chip, representing how his or her own harvest performed relative to the index. Over the course of several simulated years, participants could acquire earnings, as a result of their simulated harvests and any insurance payouts. If they were unable to repay a loan, they had to forfeit some of their productive assets, and this lowered their expected winnings in future rounds. At the end of the game, farmers received an actual cash payment proportional to their acquired winnings. In both Kenya and Peru, the researchers observed that most farmers chose to purchase some level of insurance, and also were able to infer which payout structures – in terms of thresholds of the index – were most attractive to farmers. What Carter et al. (2008) did not test, however, was the extent to which games make a measurable difference, or correlate with, participants understanding of insurance.

Similarly, Dinku et al. (2009) and Peterson and Mullally (2009) report on similar games played with farmers in Ethiopia. In this case, the researchers used games similar to those played in Peru and Kenya as part of a portfolio of communication tools to help explain index insurance to farmers. They found that most of the farmers did want to purchase insurance while playing the game. The researchers also surveyed the farmers after they had participated in the entire process that included the games as well as oral explanation of insurance contracts. Sixty-nine percent of the participants understood that insurance payouts were triggered by poor rainfall at the rainfall monitoring station, while 81% understood that if their own harvest were poor, they would nevertheless not receive an insurance payout if the rainfall at the monitoring station was above the payout threshold. These experiences using games in the context of more general attempts to explain index insurance raise an interesting question: to what extent might the use of games be a more effective strategy than traditional methods – such as written or oral explanations of index insurance contracts – at helping farmers understand insurance and develop an appropriate degree of trust? That is the central question that motivated our study.

3. Methods

We designed our game to address three issues. First, we were interested in farmers' relative preferences for three broad classes of micro-finance products: uninsured loans to finance improved seeds; insured loans (with insurance being coupled to a higher interest rate or fee) to finance improved seeds; and, insured loans to finance improved seeds, coupled to an additional insurance product offering cash payments for basic livelihood requirements. Second, we wanted to test whether farmers understood the basic concepts of index insurance well enough to be able to make an informed choice, i.e. to purchase it when it is in their interests, and not to purchase it when it is clearly not a good deal for them, such as when the risk premium is too high, or there are options to self-insure. Third, we wanted to test whether a game simulating an insurance market would be an effective communication tool, one that could be used by insurance providers to help farmers better navigate the decision of whether to purchase insurance.

3.1. Overall experimental design

To address these three issues, our experimental design thus consisted of two steps. First, it consisted of a randomized treatment. One group (the control group) received information about index insurance through what can be described as a conventional process: an expert on insurance explained how it operates to the participants, and answered any questions. Indeed, the explanation that this group received mirrored exactly the explanations that have been used in the Malawi pilot that bundled credit and index-based micro-insurance, reported on by Giné and Yang (2009) and Hochrainer et al. (2009). The other group (the treatment group) received no direct explanation about index insurance, but rather received instructions for an index insurance role-playing game, and then played that game to simulate eight consecutive years' experience. Within the treatment group, we were able to observe the participants' behavior in the form of their decisions within the game, and from this could infer their preferences among competing insurance products. We call these preferences participants' "game revealed preferences," because participants revealed them during their actions in the game.

The second step of our study was to conduct a survey of all participants, with two main parts. The first part asked participants for their preferences among the three classes of financial products (an uninsured loan, an insured loan, and an insured loan with additional livelihood insurance), compared to a fourth option, namely the absence of any micro-finance product. These preferences we can describe as "survey expressed preferences." The second part consisted of a series of questions that would test whether participants understood many of the basic concepts of how insurance and loans operate. These concepts included issues such as insurance only paying off in some years, and the basics of basis risk. With both the control and treatment groups taking the survey, we could then measure the effectiveness of both communication methods, and test whether one method produced different results from the other—either in terms of survey expressed preferences or in terms of correct answers to the test questions. Finally, as part of the survey we also asked all participants for basic demographic data, such as age and level of education. This could allow us to test whether effects observed in a real insurance market in the Malawi case also appeared in participants' game revealed and survey expressed preferences, and in their performance on the factual test questions.

We conducted the experiment in three locations in June, 2008. Two of these sites were in Ethiopia – one in Tigray and another in a rural area near Addis Ababa – where the World Food Programme and Oxfam were interested in setting up pilot insurance programs, and were able to offer logistical support, including recruitment of participants. The third location was in Malawi, near to the same location where the pilot insurance program described earlier had taken place. None of the participants in any of the locations, however, had yet participated in an actual insurance pilot program. All instructions, and the surveys, were translated into the local language.

3.2. Game design

Our purpose in this case was to design a game that would be easy enough for farmers to play, and which would simulate the core features of the four choice options. We had two versions of the game, one with basis risk and one without. All farmers started with $20 in savings. Each year they had to spend $40 for basic livelihood needs. Farmers played eight rounds of the game, with each round simulating 1 year. In all cases, the monetary figures indicated here are in "game dollars." Farmers received cash payment for participating in the game, in amounts up to about US$10 in local currency, proportionate to their final "game dollar" winnings. In each of eight rounds of the game, farmers could either use traditional seeds, or invest in improved seeds, which result in higher yields when rains are good. If they could not afford to purchase these seeds, they could choose between four options:

- No loan: farmers would not request a loan for improved seeds, and would therefore enjoy relatively low yields.
- Loan: farmers would borrow $100 to purchase higher value seeds, and would have to pay it back after the season with $10 of interest, regardless of the rains. If they have sufficient assets available to pay back the loan (from their harvest or saved from previous rounds of the game), then they have to use these to pay back the loan. If they do not have sufficient assets, then they are bankrupt, and have to stop playing.
- Protected loan: farmers again borrow $100 to purchase the same higher-value seeds. In this case, on top of the $10 interest owed at the end of the season, there is an additional fee of $22 for the insurance premium. They do not have to pay the loan back if the rains are bad (the insurer repays the loan).
- Cash-protected loan: this was the same as the protected loan, except that for an additional insurance premium of $4, farmers also purchased some livelihood insurance. The insurer would not only repay the loan but also give them $20 in cash, in the event that the rains were bad.

The yields that farmers received varied as a function of their farming inputs (i.e. seeds) and luck. In the version without basis risk, the only luck element was the rainfall. After all farmers had decided which option to choose, the game administrator rolled a six-sided die that represented rainfall as measured in a nearby weather station. If the die landed on the only side depicting a bright sun, then the rains were bad; otherwise, the five sides depicting rain indicated that the rainy season was good. In the version with basis risk, each farmer also drew a single luck card each play of the game from a stack of ten cards capturing the fact that harvest in the farmers' field may be different from what would be expected based on measured rains, such as pests, or rains in field different from weather station. One card was unlucky, eight were normal, and one card was lucky. Table 1 shows the yields that farmers received in both versions of the game, as well as their net proceeds for the year after repaying any loans due and spending $40 on annual living expenses.

We chose the payouts quite carefully. Our goal was to make sure that farmers' earnings would be higher if they took the loan than not, and that each version of insurance would come at no cost to their expected winnings. To make sure that the distributions of earnings made sense, we simulated the game using a computer model of the game, in order to calculate the probability distribution of each version. Fig. 1 shows these two distributions. As can be seen, in both versions, the potential earnings are highest in for the loan option. Taking the loan clearly dominates not taking it, in both versions. The effect of taking insurance is to tighten the distribution of earnings, including reducing the chances of going bankrupt (which in the absence of insurance is about 50%). The cash-protected loan gives the lowest probability of going bankrupt, less than 5% in the version without basis risk, and about 15% in the version with basis risk.

3.3. Survey design

We designed a survey both to elucidate participants' preferences, and to test their basic understanding of several core concepts of index insurance. The first part of the survey – eliciting participants expressed preferences – was built around a single question in which participants had to rank the four options (no loan, loan, protected loan and cash-protected loan) if they were available to them for acquiring high-yielding seeds. The second part, by contrast, was longer and more complicated. We identified fourteen principles that we felt were important to properly judge insurance contracts, and for each issue, wrote two questions. One of the questions in the pair correctly stated the principle, while the other question incorrectly stated the principle in such a way as to be completely false. Each question, then, had a correct answer,

Table 1
Cash payouts each period as a result of luck factors and choice. Note that each period farmers have top pay back the loan (with any insurance fees), unless they have insured the loan and the rains are bad. They also have to pay living expenses of 40.

Choice	Value of harvest *Net gain for year*							
Version	Without basis risk		With basis risk					
Rains	Good	Bad	Good			Bad		
Luck			L	N	U	L	N	U
No loan	50 *10*	0 *−40*	100 *60*	50 *10*	0 *−40*	50 *10*	0 *−40*	0 *−40*
Loan (cost = 110)	200 *50*	0 *−150*	250 *100*	200 *50*	150 *0*	50 *−100*	0 *−150*	0 *−150*
Protected loan (cost = 132)	200 *28*	0 *−40*	250 *78*	200 *28*	150 *−22*	50 *10*	0 *−40*	0 *−40*
Cash-protected (cost = 136)	200 *24*	0 *−20*	250 *74*	200 *24*	150 *−26*	70 *30*	20 *−20*	20 *−20*

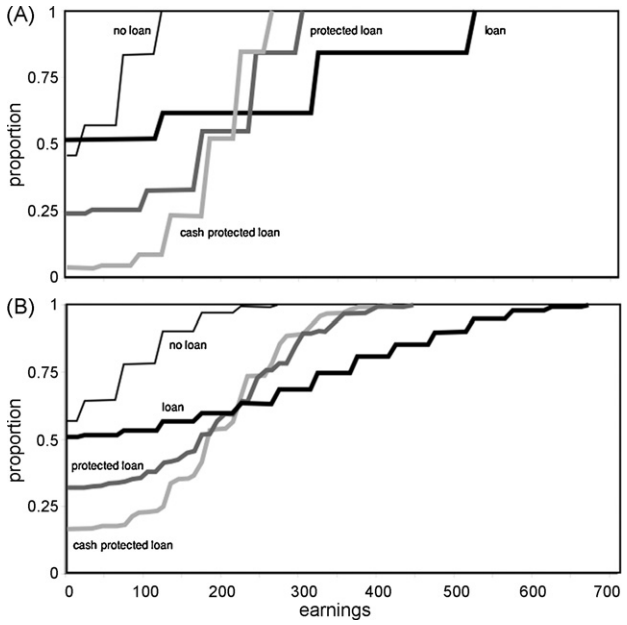


Fig. 1. Simulation of game, without (A) and with (B) basis risk. The curves measure cumulative probability, assuming a participant adopts a uniform strategy in all rounds of the game. For example, in the version without basis risk (A), one can say that for those who never take a loan, about 45% of them will go bankrupt and win nothing, 75% of them will win less than about $80, and 100% of them will win less than about $120. Insurance reduces chances of bankruptcy and tightens the distribution of winnings.

which was either true or false. We then mixed the questions up, so that the pairs of questions would not be evident to participants. Table 2 shows the list of principles, along with the true version of the two questions. It is worth noting that the interactive lecture in which the control group participated contained specific information on each of the fourteen principles, and so if members of the control treatment group were able to remember everything that they had been told, they would be able to answer all questions correctly. The treatment group learned answers to none of the fourteen sets of questions, but had to infer them from their experience with the game.

4. Results

In this section, we present three sets of data: the survey expressed preferences for insurance, from the first part of the survey; the game revealed demand for insurance, from the game playing strategies observed; and, the understanding of the core concepts of insurance. In the case of the first and third of these issues, we identify the effects of the game versus control treatment.

4.1. Survey expressed preferences for insurance

Participants expressed a clear demand to purchase insurance, with no clear preference between a basic protected loan and a protected loan with an additional cash payment to insure their livelihood security. Fig. 2 shows the relative frequency of listing each of the four options as their first, second, third, or fourth choice. A statistical test (a *multinomial logit* regression) indicated no significant difference ($p > 0.10$) between the two insurance options as the first choice.

Table 2
Part two survey principles. This table shows the fourteen principles that we tested on the true-false part of the survey. For each, we show the untranslated "true" version of the question. The "false" versions were in most cases an exact opposite of the main statement, such as, for Principle C, "Friends and neighbors can help you in times of need, and will be especially able to do so when the rains are poor.".

Code	Principle	True question version
A	Why buy insurance?	Insurance helps you when the rains are poor
B	Self-insurance options	Assets like livestock or farming equipment can help you when you have a poor harvest, as long as others in the community are doing well and can pay you a fair price form them
C	Social-capital options	Friends and neighbors can help you in times of need, but may not be able to help you when the rains are poor
D	Need to pay back loan	If you take a loan without insurance, you will have to pay it back with interest after the harvest
E	Basics of insurance loan	Insurance on a loan means that if the rains are poor, you will not have to pay back any or all of the loan
F	When a loan might be too expensive	Loans can be too expensive when the interest on the loan is more than the extra harvest that the loan would help you get
G	Basics of index insurance	Index insurance means that whether you have to pay back interest on the loan depends on what the rainfall in the community is, rather than your own harvest
H	Basics of basis risk	There is some risk with index insurance that the rains are good, and yet your own harvest is poor
J	Basis risk relative to underlying risk	The chances of having bad luck with index insurance – having to pay back the loan even though your own harvest was poor – are much smaller than the chances of having bad luck with no insurance at all
K	Fair premium costs	Insurance comes at a fair price if the premium costs about what you would receive back, averaged over many years
L	Independence of risk	If you think insurance is a good idea, then you should purchase it every year, since it is hard to tell when the rains will be good or bad
M	Figure out if the premium is above or below actuarially fair	Imagine that you live in a community where the rains are below 600 mm during the growing season about 1 year in five. Further, imagine that an insurance contract that will pay you $50 when the rains are below 600 mm during the growing season, but nothing when the rains are above 400 mm. If it costs $9 each year to purchase this insurance, and you purchase it every year, then you would expect, over many years, to receive more from the insurance company than you paid to it
N	When basis risk might be too high for you	Index insurance is risky if the rain on your own yields per hectare are sometimes much higher, and sometimes much lower, than others' in the community, even when you plant the same seed varieties
P	When a loan might be a bad deal for you	One reason to decide not to borrow money is that you think that borrowing money, and spending it on improved inputs, will not improve your harvest yields very much beyond what you would receive from normal inputs

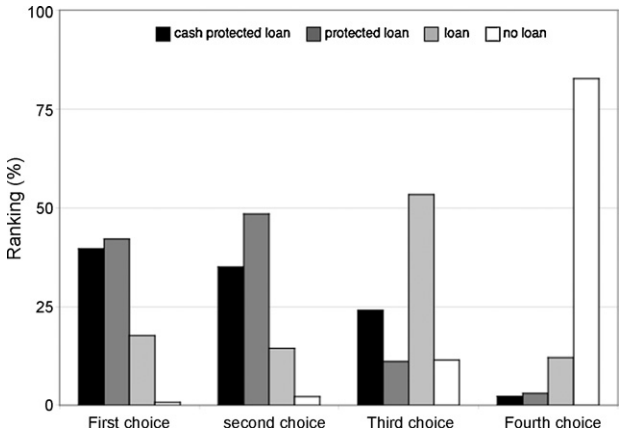


Fig. 2. Survey Part 1 answers. Respondents ranked the two insurance options more highly, either as their first or second preference. They were most likely to list the no loan option as their least favorite alternative. $n = 278$.

We then analyzed the results by group. In terms of location, we observed a greater propensity in Tigray, relative to the other two sites, towards choosing the unprotected loan. A *multinomial logit* regression indicates that the difference in the distributions between the three locations is significant ($p < 0.05$). A likely explanation can be found in the literacy rates. Due to an error in survey administration, we did not collect data on literacy in Addis, but in Tigray and Malawi we did, and we observed two things. First, there was a significant difference in the literacy rates between the two samples (X^2 test, $p < 0.01$), with 92% of the Malawi sample but only 45% of the Tigray sample being literate. Second, illiterate people were more likely (significance tested with *multinomial logit* and X^2, $p < 0.01$) to choose the loan option without insurance (34%), relative to illiterate people (20%). So the driver of the difference observed in Tigray, at least compared to Malawi, was likely literacy, and the effects of literacy are consistent with the results reported in Giné and Yang (2009). We also examined any possible relationships between survey expressed preferences and the other demographic variables that we had collected. These included gender, age, assets, land holdings, whether the person was a household head, and household size. For all of these, there were no significant correlations (*multinomial logit*, $p > 0.10$ in all cases).

The other difference that we were interested in was between the control and treatment group, which would indicate whether the experience with the role-playing game increased the attractiveness of insurance. We observed very little difference, and a *multinomial logit* regression indicated that the treatment group was not a significant ($p > 0.10$) predictor of the survey expressed preferences.

4.2. Game revealed preferences for insurance

The second issue is the game revealed preferences for insurance. For this we examined the data from the plays of the game. Fig. 3 shows the results from all treatment group participants, with the top graph (A) aggregating the groups that played the versions with and without basis risk, and the bottom graph showing the more complicated disaggregated results. With respect to the top graph, there are two points worth making. First, the results track quite closely, at an aggregate level, the survey expressed preferences. There was a clear preference for the two choices that included insurance, and a clear disfavor over the choice of no loan. Indeed, only one participant chose not to take a loan, in two of the eight rounds of the game. Second, there was a slight increase in the number of people who chose the unprotected

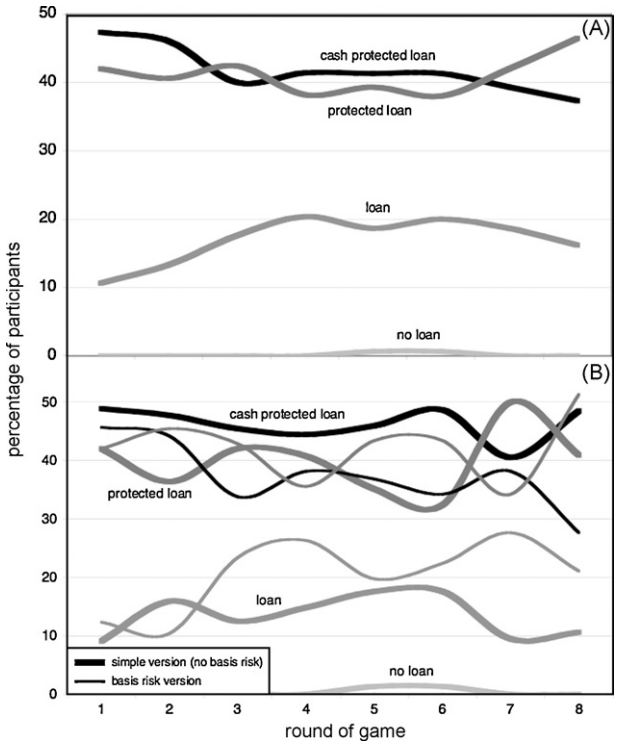


Fig. 3. Aggregate play of the game. Each curve shows the percentage of participants opting for each of the four options, over the course of the eight rounds of the simulation game. The two insurance options were consistently more popular than the other two options. The top graph (A) shows the aggregate results, while the bottom graphs (B) shows these for the separate groups playing with and without basis risk. $n = 168$.

loan as the game progressed, falling again slightly at the end. The first of these is consistent with rational choice: after several rounds of the game, most players had saved enough money so as to be able to avoid going bankrupt, making it more attractive to choose the unprotected loan. The latter is consistent with behavioral economics theory: it may be that an "endowment effect" (Kahneman et al., 1990; Kahneman and Tversky, 1979) had set in as players amassed earnings, and they did not want to lose that money as the end of the game drew near, choosing the insurance option.

The lower graph (B) disaggregates the results into the equally sized groups playing with and without basis risk. We have not been able to test differences for statistical significance, but can still observe a few trends. First, among the group playing with basis risk (narrow lines), the black and dark grey curves (black representing cash protected loan; dark grey representing protected loan) continue to intersect. This is in contrast to the group playing without basis risk, where the cash-protected loan, with the exception of round 7, remains above the protected loan. One explanation is that the more complicated of the two choices (cash-protected loan) loses some of its attractiveness in the more complicated game (basis risk): players may have been simply overwhelmed by details when playing with basis risk, and so choose the simpler insurance version. Second, the loan with basis risk is generally above the loan without basis risk. This indicates a slightly higher preference for no insurance (or a lower preference for insurance) in the presence of basis risk. Put more simply: the presence of basis risk made insurance slightly less attractive, compared to taking the uninsured loan.

The final piece of analysis was to compare game revealed and survey expressed preferences. At the aggregate level they appear comparable, but what about at the individual level? If they are

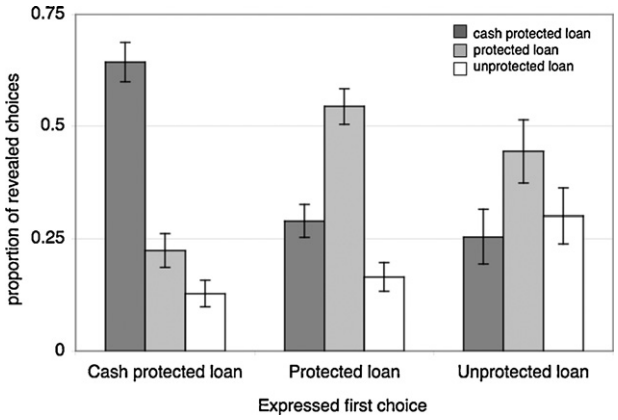


Fig. 4. Relationship between survey expressed and game revealed preferences. Each block of bars represents a group of participants that expressed a first choice preference for one of the three micro-finance options. Within each block, the three bars represent the relative frequency with which members of this group chose each of the three options during the simulation game. Error bars indicate one standard error. $n = 168$.

comparable at the individual level, then one can be more confident that answers on a survey would predict game behavior, and vice versa, and ultimately more confident that survey answers would correlate with actual market behavior. To analyze this, we sorted participants into three groups, based on their first choice in Part 1 of the survey (among those who had played the game, there were only two people who had indicated that their first choice was no loan, and so we did not evaluate this group). For each group, we then calculated the proportion with which they had chosen each of the three options during the eight rounds of the game. Fig. 4 shows the results. Those who expressed a first choice preference for the cash-protected loan chose the cash-protected loan 64% of the time while playing the game, compared to the protected loan 22% and the unprotected loan 13%. Likewise, those who indicated a first choice preference for the protected loan chose this option most frequently: 54% compared to 29% and 17%. For each of these two groups, there was a significant difference ($p < 0.05$) between the most frequently played strategy and either of the other strategies. Those who expressed a first choice preference for the unprotected loan, however, did not reveal a preference for this during the game, and indeed there are no significant differences ($p > 0.05$) between the rate with which any of the three strategies were played. Hence, an expressed preference for either the cash-protected loan or the protected loan was correlated with a game revealed preference for the same option, but a survey expressed preference for the unprotected loan was not correlated with a clear game revealed preference for any of the three options.

4.3. Understanding of core concepts about insurance

The final question was whether people understood the basic concepts of insurance, as uncovered by their answers on the second part of the survey. Here it was again possible to measure whether there were differences in understanding as a function of being in the control or game treatment groups, and whether there was a correlation between understanding and either the game revealed or survey expressed preferences. Fig. 5 the proportion answering questions for each concept correctly, disaggregated into control and treatment groups. For ten of the fourteen principles, participants did significantly better than chance (i.e. 50% correct) in both control and treatment groups. For principle *F* (when a loan might be too expensive), participants' answers were significantly worse than chance, whereas for principles *K* (what is a fair premium for insurance), members of the game group did

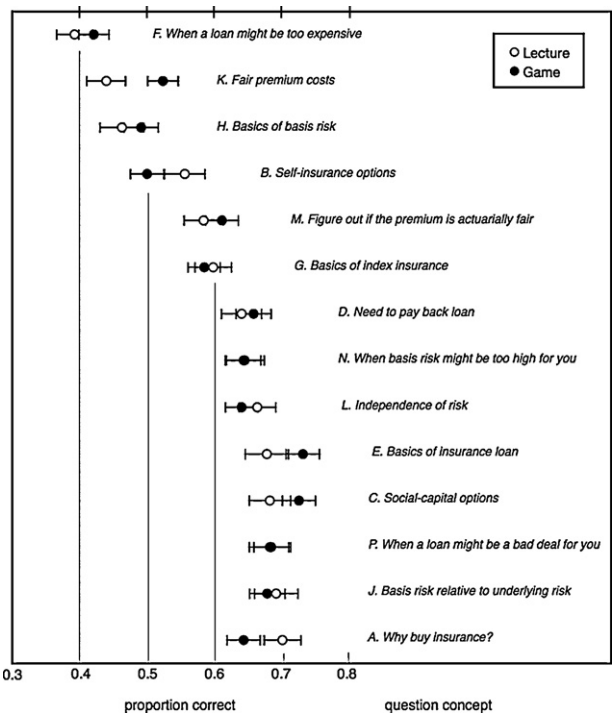


Fig. 5. Proportion of correct answers for each of the concepts in part two of the survey. Error bars indicate one standard error. The only significant differences between the two treatment groups were for concepts *K* ($p < 0.05$) and *A* ($p < 0.10$). $n = 278$.

significantly better ($p < 0.05$) than the control group (a majority of the game group answered correctly, but a majority of the control group answered incorrectly). This may indicate that under normal circumstances farmers tend to perceive the cost of financial services as being excessive, but after simulating the insurance market they understand that an insurance premium can be fair. For one of the principles – A, the basic insurance concept – people who had learned this concept in the control group did significantly better ($p < 0.10$) than the game group. There was significant ($p < 0.01$) correlation between literacy and test scores, with literate people performing better than illiterate. There were no other significant correlations between test scores and demographic variables.

Did people's understanding of the insurance concepts correlate with their game revealed or survey expressed preferences? We found a significant correlation ($p < 0.05$) with both, with the

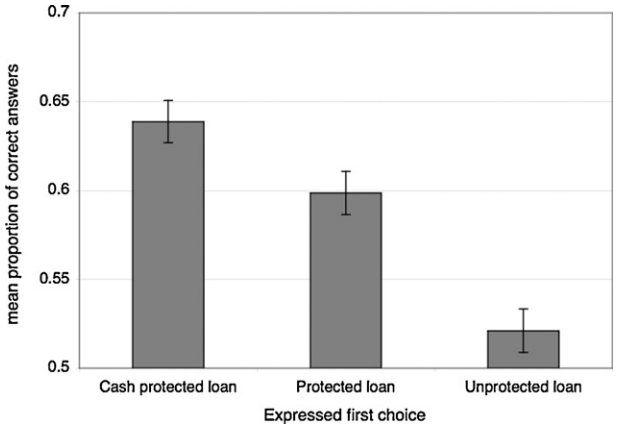


Fig. 6. Average test scores for groups defined by their survey expressed first choice preference. Error bars indicate one standard error. $n = 278$.

fraction of questions answered correctly being positively correlated with both a game revealed and survey expressed preference for the cash-protected loan, and a negative correlation with a game revealed and survey expressed preference for the unprotected loan. Fig. 6 shows the average test scores for those who expressed a first choice preference for each of the three finance options.

5. Discussion

The results provide answers to our three research questions with varying levels of clarity. Our first question concerned the extent to which farmers understand crop insurance as a means to reduce the risk associated with climate variability. The results from the set of test questions suggest that many farmers, even after learning about index insurance through a simulation game or a more conventional education session, do not understand many of the core concepts of insurance that would be necessary to make a fully informed and educated choice. Their level of understanding correlates significantly with education, suggesting that additional training may be necessary to familiarize lesser-educated farmers with insurance.

Our second question concerned whether a communication strategy built around the use of simulation games improves understanding more than more conventional approach of simply explaining how insurance works. Here, our results suggest that the use of a role-playing game does not out-perform traditional explanations, with one important exception being the perception of fairness in premium costs, where the game group significantly out-performed the control group. Indeed there were almost no observed differences between the two strategies, but there are two caveats. The first is to note that our results could well be an artifact of both the explanation used and the game deployed. In two of the locations, for example, factors largely outside out control may have put game participants at a disadvantage. In both Malawi and Tigray, the role of the six-sided die produced 8 consecutive years of good rainfall for one of the two game groups, and while this added to participants' winnings, it did not give them an opportunity to experience an insurance payout (meant to be the main way for them to learn about the insurance mechanism). In Tigray, a short intense rainfall on the tin-roofed building we were in made it very difficult to hear the directions for the game, something that we were not completely aware of until the game ended. The second caveat is to note that game participants did do just as well as the control group, even though the control group was given clear information that would enable them to answer all of the test questions correctly. Additionally, according to one of the most experienced agricultural extension officers involved in the marketing of the actual Malawi pilot, who contributed to implementing this research: "the game makes it much easier for us to explain to farmers how insurance really works." This should be a sign of encouragement that simulation games could be very useful, even if our fieldwork we did not observe them to be more useful than traditional explaining, which is what we had hoped to find when designing the study.

Our third question concerned the extent to which farmers' desire to purchase insurance correlate with their level of understanding. Here we observed a clear correlation: farmers who better understood how insurance operated were more likely to express a preference for it, and more likely to reveal a preference for it in how they behaved in the role-playing game. These results show up in two ways. First, they appear in the correlation between participants' test scored and their preference for insurance. Second, they appear in the differences in game revealed preference for more complicated insurance products depending on whether those products entailed basis risk. Participants having to decide between options in the presence of basis risk were more likely to opt for simpler options, ones that they could better understand.

All of these answers suggest that there is a need for continued attention to how well farmers understand index insurance, as part of pilot projects and programs to introduce it to small-holder communities in Africa. The use of simulation games may be a valuable part of such efforts, although our results do not demonstrate that they are necessarily superior to more conventional approaches towards education.

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